

3.1.5 Common Built-in Functions (min, max, len, sum)

len () عناصر گنتا ہے، min () سب سے چھوٹا، max () سب سے بڑا، sum () اعداد کا مجموعہ، اور sorted () نئی ترتیب شدہ

فہر...

CS XII

Chapter 3: Programming Fundamentals

Lecture 06 of 18

3.1.5 Common Built-in Functions (min, max, len, sum)

LEARN IT

Several built-in functions work across lists, tuples, sets and dictionaries. You do not call them with a dot on the object (except when a type also has its own method, such as `list.sort()`).

Function	Purpose	Works on
<code>min()</code>	Smallest element (numeric or alphabetical)	List, tuple, set, dictionary keys
<code>max()</code>	Largest element	List, tuple, set, dictionary keys
<code>len()</code>	Number of elements (for a dict: number of keys)	List, tuple, set, dictionary
<code>sum()</code>	Adds numeric elements	List, tuple, set, dictionary values via <code>sum(d.values())</code>
<code>sorted()</code>	Returns a new sorted list ; original unchanged	List, tuple, set, dictionary keys

```
s = {5, 1, 9, 3, 7}
print(len(s))
print(min(s))
print(max(s))
print(sum(s))
print(sorted(s))
```

OUTPUT

```
5
1
9
25
[1, 3, 5, 7, 9]
```

$5 + 1 + 9 + 3 + 7 = 25$. `sorted(s)` always returns a **list**, even when the input is a set.

WORKED EXAMPLE

The same five functions on four structures.

```
nums = [8, 3, 5, 3]
t = (8, 3, 5, 3)
s = {8, 3, 5}
d = {"math": 80, "eng": 70, "cs": 90}
print(len(nums), min(nums), max(nums), sum(nums), sorted(nums))
print(len(t), min(t), max(t), sum(t), sorted(t))
print(len(s), min(s), max(s), sum(s), sorted(s))
print(len(d), min(d), max(d), sum(d.values()), sorted(d))
```

OUTPUT

```
4 3 8 19 [3, 3, 5, 8]
4 3 8 19 [3, 3, 5, 8]
3 3 8 16 [3, 5, 8]
3 cs math 240 ['cs', 'eng', 'math']
```

On the dictionary, `min(d)` and `max(d)` compare **keys as strings**: "cs" comes first alphabetically, "math" last. `sum(d)` would try to add the keys "math" + "eng" + ... and fail. Use `sum(d.values())` — 80 + 70 + 90 = 240. The set dropped the duplicate 3, so `len(s)` is 3 and `sum(s)` is 16.

`sorted()` versus `.sort()`

```
original = [8, 3, 5]
print("before sort:", original)
print("sorted():", sorted(original))
print("after sorted():", original)
original.sort()
print("after .sort():", original)
```

OUTPUT

```
before sort: [8, 3, 5]
sorted(): [3, 5, 8]
after sorted(): [8, 3, 5]
after .sort(): [3, 5, 8]
```

`.sort()` exists only on lists and rewrites the original. `sorted()` works on any iterable and leaves the original alone.

REMEMBER IT

LEN-MIN-MAX-SUM — count, smallest, largest, total. For a dictionary: `len` / `min` / `max` / `sorted` use **keys**; `sum` needs `.values()`.

COMMON MISTAKE

`sum()` needs numbers. `sum(["a", "b"])` raises `TypeError`. Empty structures: `len([])` is 0, but `min([])` and `max([])` raise `ValueError` because there is no smallest of nothing.

اردو میں سمجھیں

`len()` عناصر گنتا ہے، `min()` سب سے چھوٹا، `max()` سب سے بڑا، `sum()` اعداد کا مجموعہ، اور `sorted()` نئی ترتیب شدہ فہرست دیتا ہے بغیر اصل کو بدلے ہیں۔ ڈکشنری پر `min` اور `max` کلیدوں پر چلتے ہیں؛ قدروں کا مجموعہ `sum(d.values())` سے نکالیں۔ `sort()` صرف فہرست کا میٹھڈ ہے جو اصل فہرست بدل دیتا ہے۔

CHECK YOURSELF

1. For `s = {5, 1, 9, 3, 7}`, what are `len(s)`, `min(s)`, `max(s)`, `sum(s)`?
2. Does `sorted(s)` return a set or a list?
3. How do you total the values of `{"math": 80, "eng": 70, "cs": 90}`?

Answers: 1. 5, 1, 9, 25. 2. A list: `[1, 3, 5, 7, 9]`. 3. `sum(d.values())` → 240.

Next lecture: 3.1.6 Activity: Student Attendance System